

Political Bias and Temporal Dynamics in Large Language Models

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Abstract

Large Language Models (LLMs) are increasingly deployed as intermediaries for accessing, interpreting, and evaluating political information. Prior work has demonstrated that LLMs exhibit systematic political biases, reflected in ideological alignment, sentiment toward political actors, and framing choices. However, most existing studies treat political bias as a static property, evaluating models at a single point in time. This assumption overlooks the rapid evolution of LLMs, which undergo frequent updates to training data, safety mechanisms, and alignment procedures.

In this work, we investigate political bias as a temporal phenomenon, analyzing how political attitudes expressed by LLMs evolve across model generations. Focusing on longitudinal comparisons within model families, we examine how changes in safety data distributions and alignment interventions correspond to shifts in political stance and sentiment. Our study integrates Prediction Inconsistency (IC) and sentiment label distributions to provide a dynamic perspective on value alignment in large language models.

1 Introduction

Large Language Models (LLMs) have become central intermediaries in how individuals access, interpret, and evaluate political information. Their widespread use in tasks such as summarization, explanation, sentiment analysis, and question answering grants them substantial influence over contemporary digital public spheres. As these systems are increasingly embedded in search engines, social media platforms, and decision-support tools, their latent political orientations carry meaningful societal and political consequences.

A growing body of evidence suggests that LLMs are not ideologically neutral. Instead, political tendencies emerge from architectural decisions, training data composition, and post-training alignment strategies [Santurkar et al., 2023, Buyl et al., 2026, Hartmann et al., 2023, Bang et al., 2024]. When models systematically favor or disfavor particular political ideologies, policies, or actors, they may function as implicit instruments of digital soft power, especially when deployed at scale across languages and geopolitical contexts.

Empirical research further shows that political bias in LLMs is not limited to isolated prompts but constitutes a structural property of model behavior. Studies have identified consistent ideological patterns across economic and social dimensions, as well as systematic sentiment differences toward political entities [Hartmann et al., 2023, Bang et al., 2024]. Moreover, LLMs may exhibit bias amplification, whereby biases present in training data are reproduced or intensified in generated outputs, reinforcing existing societal asymmetries [Zhu et al., 2024].

Despite this progress, most prior work evaluates political bias under a static assumption, treating models as fixed artifacts. This perspective is increasingly inadequate given the rapid pace of LLM development. Model families such as LLaMA undergo substantial changes across generations, including modifications to training corpora, safety data distributions, and reinforcement learning-based alignment procedures [Touvron et al., 2023, Dubey et al., 2024]. These interventions plausibly affect not only factual reliability and safety but also political framing and sentiment.

1.1 Research Questions and Contributions

To address this gap, we study political bias through a temporal and longitudinal lens. Specifically, this paper addresses the following research questions:

- **RQ1:** How does political bias in LLM outputs change across successive model generations?
- **RQ2:** To what extent do safety alignment and red-teaming interventions correspond to shifts in political stance and sentiment?
- **RQ3:** Are temporal changes in political bias uniform across ideological dimensions and political entities, or do they exhibit asymmetric patterns?

Our contributions are threefold:

- We conceptualize political bias in LLMs as a dynamic property, rather than a static characteristic.
- We provide a longitudinal analysis of political bias across model generations using complementary stance- and sentiment-based metrics (IC and label distributions).
- We empirically link observed bias shifts to documented alignment and safety interventions, contributing to a deeper understanding of value alignment over time.

2 Related Work

Research on political bias in large language models spans ideological measurement, sentiment analysis, temporal evaluation, and mitigation. We review the most relevant work along three axes.

2.1 Measuring Political Bias in Language Models

Early studies mapped LLM outputs onto established ideological frameworks, demonstrating that conversational models express coherent political orientations. Some argue that LLMs reflect the aggregate opinions embedded in their training data [Santurkar et al., 2023], while others situate ChatGPT along a left-libertarian axis using political compass evaluations [Hartmann et al., 2023]. Subsequent work refined bias measurement by distinguishing between content and style. A two-pronged framework analyzing both explicit stance (“what is said”) and framing or lexical polarity (“how it is said”) reveals subtle partisan effects even in ostensibly neutral responses [Bang et al., 2024]. Target-oriented sentiment classification further operationalizes political bias by measuring sentiment toward specific political actors or parties [Wang et al., 2025, Chen et al., 2024].

2.2 Temporal Dynamics and Model Evolution

Only recently has attention shifted toward the temporal dimension of LLM behavior. Research shows that LLMs exhibit temporal generalization failures, with performance and biases drifting as political and informational contexts evolve [Zhu et al., 2024]. Models implicitly encode the ideological preferences of their creators, suggesting that updates to training and alignment pipelines may systematically reshape political outputs [Buyl et al., 2026]. Technical reports on the LLaMA family provide concrete evidence of such evolution. Between LLaMA 1, 2, and 3, Meta introduced substantial changes to safety data distributions, reinforcement learning objectives, and red-teaming protocols [Touvron et al., 2023, Dubey et al., 2024]. Comparative evaluations indicate that newer models handle politically nuanced and ambiguous content differently, particularly in borderline factual or normative cases [Crum et al., 2024].

2.3 Mitigation and Alignment Interventions

A parallel line of work investigates methods for mitigating political bias. Reinforced calibration techniques have been proposed to reduce ideological skew [Banko et al., 2025], while other research demonstrates that political stance is encoded in internal model activations and can be selectively steered [Banko et al., 2025]. These findings suggest that political bias is not merely emergent but amenable to targeted intervention. However, mitigation introduces trade-offs. Alignment and safety guardrails may reduce expressive or argumentative capacity, potentially altering political nuance [Bonaldi et al., 2024]. The LLaMA 3 report explicitly frames safety alignment and red-teaming as central design goals, raising questions about how such interventions affect political expression over time [Dubey et al., 2024].

3 Methodology

3.1 Measuring Political Bias of LLMs

To evaluate the evolution of political bias, we adapt the framework proposed by El Bouanani et al. [2025], which quantifies bias through target-oriented sentiment classification (TSC). While their work focuses on a static snapshot of models, we extend this methodology to a temporal dimension to analyze how these biases shift over time.

3.1.1 Data Selection

Entity Selection: We construct a set of 250 political figures sampled from major geopolitical regions to ensure broad ideological and demographic coverage. The selected entities span the full political spectrum from far-left to far-right and are diversified across individual attributes, including gender, ethnicity, socio-economic background, and age. This diversity is intended to reduce confounding effects caused by over-representation of specific political or demographic groups and to enable a fine-grained analysis of entity-related bias.

Sentence Selection: We select 60 sentences from news-style political discourse, categorized into positive, negative, and neutral sentiments. Consistent with the criteria in El Bouanani et al. [2025], we exclude sentences containing explicit role-specific references (e.g., “the current U.S. president”) to ensure that the political figure’s name is the only varying component, thus maintaining counterfactual consistency during entity substitution.

Examples:

- Positive: X contributed to easing tensions during the negotiations.
- Negative: X was criticized for mishandling the crisis.
- Neutral: X addressed the issue during a public speech.

3.1.2 Metrics Explanation

We operationalize political bias as prediction inconsistency (IC), a metric defined by El Bouanani et al. [2025] to measure the instability of sentiment predictions when target entities are swapped. For a given sentence, a politically unbiased model is expected to produce identical sentiment predictions regardless of the political entity being substituted. Deviations from this behavior indicate entity-related bias.

We quantify this phenomenon using IC, defined as the average entropy of sentiment predictions across entities for each sentence. Higher entropy values correspond to greater instability in predictions and, consequently, stronger political bias. To calculate IC, we first perform entity replacement for each sentence template to generate a prediction set. We then derive the probability distribution for each sentiment label $l \in \{\text{positive, neutral, negative}\}$ using:

$$P(l|s) = \frac{|\{e \in E : \text{pred}(s, e) = l\}|}{|E|} \quad (1)$$

where E is the set of all entities, s is the sentence template, and $\text{pred}(s, e)$ returns the model’s predicted sentiment label for sentence s with entity e .

These probabilities are used to compute the Shannon entropy for a specific sentence via the formula:

$$H(s) = - \sum_{l \in L} P(l|s) \log P(l|s) \quad (2)$$

where $L = \{\text{positive, neutral, negative}\}$. Finally, the global bias score is determined by averaging the entropy values across all m sentences in the experiment:

$$\text{IC} = \frac{1}{m} \sum_{i=1}^m H(s_i) \quad (3)$$

where an IC value of 0 represents a perfectly unbiased and consistent model.

3.1.3 Experiment Setup

We employ a 9-shot prompting strategy, providing nine labeled sentiment classification examples before each query. This few-shot setting is chosen to stabilize model behavior and reduce variability caused by prompt underspecification. To further constrain the output space, we explicitly instruct the model to produce one of three labels only: positive, neutral, or negative. Consistent with the recommendations of Röttger et al. [2024], we acknowledge that LLM evaluations of values and opinions are highly sensitive to prompt constraints. To ensure the robustness of our results across the time dimension, we incorporate extensive robustness tests as suggested, making local rather than global claims about the manifested political biases to avoid overgeneralization.

3.2 Measuring Political Bias Over Time

3.2.1 Research Objective and Extension Logic

While Section 3.1 establishes a framework for measuring political bias through Target-Oriented Sentiment Classification (TSC) in a static context, this section extends the analysis to the temporal dimension. We treat Large Language Models (LLMs) as evolving systems rather than static tools. The primary objective is to determine whether political bias—defined as Prediction Inconsistency (IC)—is being mitigated or amplified as model families transition through successive generations and alignment phases. To ensure experimental control, this section reuses the 250 entities, 60 sentence templates, and prompts defined in the previous section. By keeping these constants, we isolate the Model Version as the primary independent variable, allowing for a pure measurement of how model evolution affects ideological consistency.

3.2.2 Longitudinal Model Selection and Lineages

To capture a representative trajectory of AI evolution, we categorize models into “Lineages.” This selection process allows us to observe bias across three distinct evolutionary paths:

- **Generational Scaling:** tracking architectures across releases (e.g., ChatGPT gpt-3.5-turbo through gpt-4o).
- **Alignment Progression:** comparing Base vs. Instruct (RLHF-tuned) models within the same family (Llama 3.2).
- **Cross-family comparison:** including OpenAI (ChatGPT) and Meta (Llama) families.

3.2.3 Experimental Procedure and Implementation

The experiment follows a deterministic execution plan: (1) Deterministic Sampling: Temperature = 0 for all queries. (2) Snapshot Testing: Versions tested on identical prompt sets. (3) IC and label-distribution computation per model. (4) Comparative attribution: results grouped by family and alignment. Results are visualized using trajectory plots where the x-axis represents model version and the y-axis represents IC score; divergence analysis (trajectories moving upward indicate models grow more opinionated) and stability analysis (flat IC suggests political bias originates from pre-training and resists alignment tuning) are reported.

4 Results

Figure 1 summarizes IC across versions. We report IC and label distributions below.

4.1 ChatGPT

IC is near zero for **gpt-3.5-turbo**. It jumps to ≈ 0.67 for **gpt-4** (the highest among all versions) and then plateaus at ≈ 0.61 for **gpt-4-turbo**, **gpt-4o-mini**, and **gpt-4o** (overall mean 0.501). Thus the main increase in political bias, as measured by IC, occurs at the transition from gpt-3.5-turbo to gpt-4; later gpt-4-family models maintain high but slightly lower inconsistency.

Label distributions shift sharply. Gpt-3.5-turbo assigns **100% positive**; no neutral, negative, or UNKNOWN. Gpt-4 gives $\approx 60\%$ positive and $\approx 40\%$ UNKNOWN. Gpt-4-turbo, gpt-4o-mini, and gpt-4o each give $\approx 30\%$ positive and $\approx 70\%$ UNKNOWN. None of the models output neutral or negative in this evaluation. The rise in UNKNOWN in the gpt-4 family may reflect greater refusal or hedging, while still producing high inconsistency when the entity is varied.

4.2 Llama

Pre-trained (Base): IC decreases with size: 1B ≈ 0.178 , 3B ≈ 0.118 . **Post-trained (Instruct):** 1B has IC ≈ 0.028 (much lower than Base); 3B remains ≈ 0.120 (similar to Base); vision 11B has the lowest IC (≈ 0.005). Instruction tuning therefore strongly reduces IC for the 1B model and yields minimal inconsistency for the 11B vision model; the 3B model is largely unchanged by alignment.

Label distributions differ by size and stage. Base 1B/3B skew negative; Instruct 1B becomes $\approx 97\text{--}98\%$ negative. Instruct 3B shows more positive and neutral (e.g., $\approx 52\%$ positive, 10% neutral, 36% negative); Instruct vision 11B is more balanced ($\approx 32\%$ positive, 30% neutral, 34% negative). Alignment thus shifts sentiment distributions in a size-dependent way.

4.3 Cross-family Comparison

ChatGPT exhibits higher IC (0.61–0.67) than the Llama models we evaluated (0.005–0.178). ChatGPT outputs are dominated by positive and UNKNOWN; Llama models produce substantial negative and neutral shares. Political bias, as captured by IC and label distribution, therefore varies by family, version, and alignment.

5 Discussion

Our results show that political bias in LLMs is not static. For ChatGPT, IC rises sharply with gpt-4 and then plateaus; the shift from all-positive to large UNKNOWN shares may reflect different refusal or classification behavior while still yielding high inconsistency. For Llama, instruction tuning reduces IC for the 1B and 11B vision models but not for 3B; sentiment distributions change markedly with alignment and size. Cross-family, ChatGPT has higher IC and different label patterns than Llama, underscoring that bias is family- and alignment-dependent.

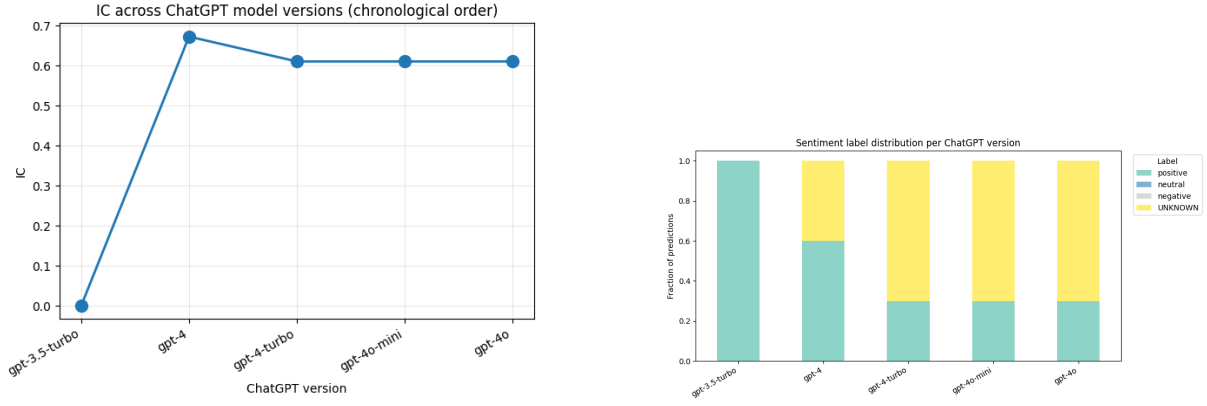


Figure 1: Left: IC across ChatGPT versions (chronological). Right: IC for Llama 3.2 Base vs. Instruct.

Limitations. We evaluate two families and a fixed set of versions; results may not generalize. IC captures prediction inconsistency under entity substitution but not other dimensions of bias (e.g., framing or ideological stance). The high share of UNKNOWN for some ChatGPT versions complicates interpretation. Differences in API behavior and model access may confound cross-model comparison.

Future work. Extend to more families, longer time horizons, and other bias metrics; relate IC to fairness and stance measures.

6 Conclusion

This work presents a temporal analysis of political bias in Large Language Models, extending static bias measurement frameworks to capture the dynamic nature of model evolution. By tracking Prediction Inconsistency (IC) and label distributions across model generations and comparing Base vs. Instruct variants within the Llama family and across ChatGPT versions, we provide insights into how political bias evolves over time. Our findings suggest that political bias, as measured by IC, exhibits strong variation across model families and versions: ChatGPT shows a sharp rise in IC at the gpt-4 transition and high UNKNOWN shares; Llama shows size- and alignment-dependent patterns. We argue for longitudinal evaluation in bias research and for reporting both IC and label distributions when comparing models.

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